PRISM surveillance report

PatientID_0003_Timepoint_1 | scan day 90 from diagnosis | early post-RT (0-84 days after the end of radiotherapy)

Risk of documented progression within 120 days

0.46 (95% interval 0.26-0.71)

Tier: middle third of development-cohort predictions (cut points 0.26 / 0.50). 80% conformal set: both outcomes (progression and no progression).

Referral flags

[!] early post-RT window: scan 25 days after the end of RT, inside the 84-day early post-RT window where treatment effects can mimic progression

[!] out of distribution: MGMT status unknown

[!] uncertain: the 80% conformal prediction set contains both outcomes

Action: flagged for clinician review

Contributions (exact additive log-odds; positive = towards higher risk)

Intercept -0.39 clinical +0.14 timing -0.34 treatment +0.43 = logit -0.17. Imaging: not part of this model.

-0.27 MGMT status unknown: yes (clinical)
-0.27 scan number = 1 (timing)
+0.16 RT started: yes (treatment)
+0.16 male sex: no (clinical)
+0.15 previous brain tumour: no (treatment)

Inputs as of the scan day (source: MU clinical workbook)

Age 57, female; IDH wildtype; MGMT unknown; RT started yes, completed yes; days since RT end 25; chemotherapy started yes. All 23 inputs known by the scan day: yes (each with source column and as-of day in the record).

Volumetric change (indicative, not RANO)

Enhancing 33.8 cc; FLAIR hyperintensity 44.3 cc.

First scan: no prior or nadir.

Summary (local LLM; every sentence cites record fields; checked deterministically)

FINDINGS: This scan is 90 days from diagnosis, 25 days after radiotherapy ended, in the early post-RT phase. In this window, treatment-related change can look like progression on MRI, which the model cannot separate. The model estimates a 46% (95% interval 26-71%) risk of documented progression within 120 days. This is in the middle third of the model's estimates on its training scans, above the average. The largest contributions are MGMT status unknown and scan number one, both towards lower risk. The model did not use the MRI images or tumour volumes, steroid dose, or symptoms. The scan is flagged for clinician review due to the early post-RT window, out of distribution, and uncertain status.

IMPRESSION: The model estimates a 46% risk, in the middle third above the average, with active flags for clinician review.

Faithfulness check: 8/8 claims supported; active flags mentioned: all; forbidden content: none.

Limitations

Model discrimination in development cross-validation AUC 0.70 (single institution, MU-Glioma-Post); the risk is the probability of documented progression within 120 days, not a diagnosis of progression or pseudoprogression; volumes are indicative, not a RANO assessment.